Specification

ZR/YC-0022-04 A2

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AC1250V 250-800A

CE RoHS COMPLIANT

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AC1250V 250-800A

CE RoHS COMPLIANT

Basics

- Conform to: IEC 60269
- Rated Voltage: AC1250V(-D/DN/DT Rated Voltage: AC1250V/DC1000V（UL）)
- Rated Current: 250A～800A
- Utilization Category: aR
- Breaking Capacity: AC 100kA(DC 50kA: time constant 10ms)
- Conform to IEC60269 /UL248
- RoHS, CE Compliant

Features in low I²t, high current limitation, high breaking capacity. Suitable for rail transport, electric vehicles, ships and other industrial applications. Used for energy storage system, power system, conductor, and short - circuit overcurrent devices and equipment, as well as backup protection.

Note: All test data are measured in a DC environment, test parameters and connector in accordance with IEC 60269.

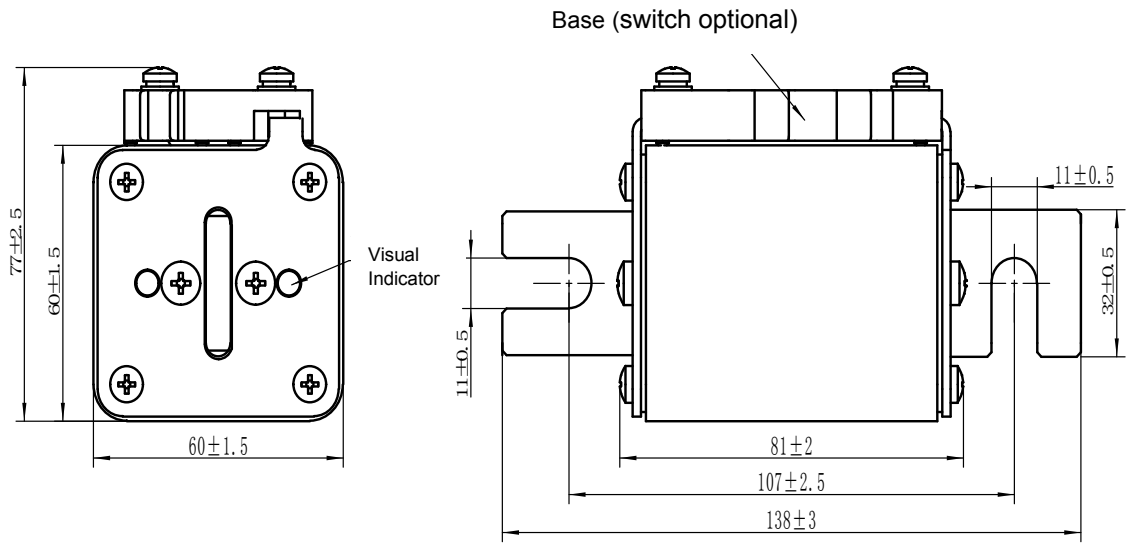
	Part Number	Size	Current (A)	I²t (A^2s)		Loss (W)		Weight (g)	Min /Max Package (pcs)	Mounting
				Prearc	Clearing	$0.5I_n$	I_n			
1	RS306-2-T5Z-250A1250V	2	250	9000	61000	12	72	910×（1±3%）	2/28	Bolt M10 Torque 32±1N·m
2	RS306-2-T5Z-315A1250V		315	17000	112000	14.5	85			
3	RS306-2-T5Z-350A1250V		350	22000	150000	16.5	92			
4	RS306-2-T5Z-400A1250V		400	32000	210000	19	100			
5	RS306-2-T5Z-450A1250V		450	45000	298000	21	108			
6	RS306-2-T5Z-500A1250V		500	61000	402000	23.5	115			
7	RS306-2-T5Z-550A1250V		550	80000	521000	26	122			
8	RS306-2-T5Z-630A1250V		630	114000	744000	30	135			
9	RS306-2-T5Z-700A1250V		700	160000	1043000	32.5	141			
10	RS306-2-T5Z-800A1250V		800	240000	1562000	36	155			

Note: 1. If end needs visual indicator, add "-T" e.g. RS306-2-T5Z-700A1250V-T

2. If no visual indicator is needed, add "-N" e.g. RS306-2-T5Z-700A1250V-N (No visual indicator and base)

Outline Dimensions (mm)

Outline Mounting



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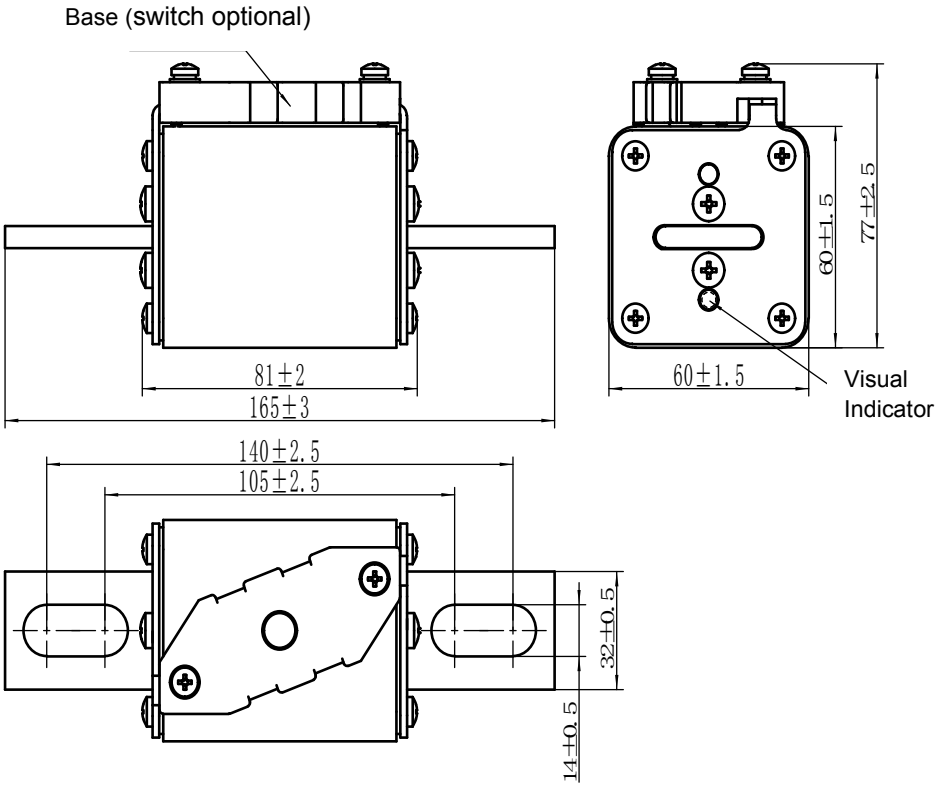
	Part Number	Size	Current (A)	I²t (A²s)		Loss (W)		Weight (g)	Min /Max Package (pcs)	Mounting
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Note: 1. if end needs visual indicator, add "-T" e.g. RS306-2-Q5Z-700A1250V-T

2. If no visual indicator is needed, add "-N" e.g. RS306-2-Q5Z-700A1250V-N (No visual indicator and base)

Outline Dimensions (mm)

Outline Mounting



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	Part Number	Size	Current (A)	I²t (A²s)		Loss (W)		Weight (g)	Min /Max Package (pcs)	Mounting
				Prearc	Clearing	$0.5I_n$	I_n			
1	RS306-2-S5P-250A1250V	2	250	9000	61000	12	72	900×（1±3%）	1 20	
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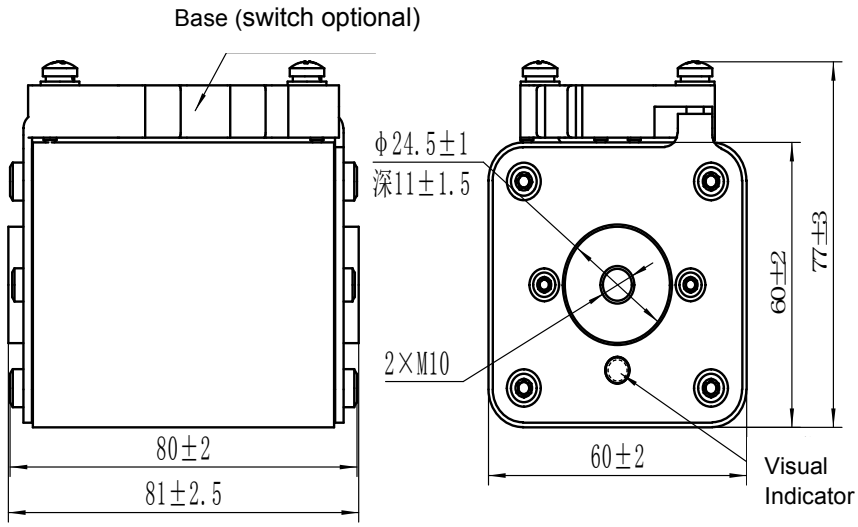
Note: 1. Above I²t measured under standard test environment; I²t test parameter: AC1260V

2. If end needs visual indicator, add "-T" e.g. RS306-2-S5P-700A1250V-T

3. If no visual indicator is needed,add "-N" e.g. RS306-2-S5P-700A1250V-N (No visual indicator and base)

Outline Dimensions (mm)

Outline Mounting



RS306-2-S5P Mounting

a）When installing, the screw-in size of the stud should not exceed 11mm (Recommend 10mm);

b）When tightening the nut, keep the stud from rotating;

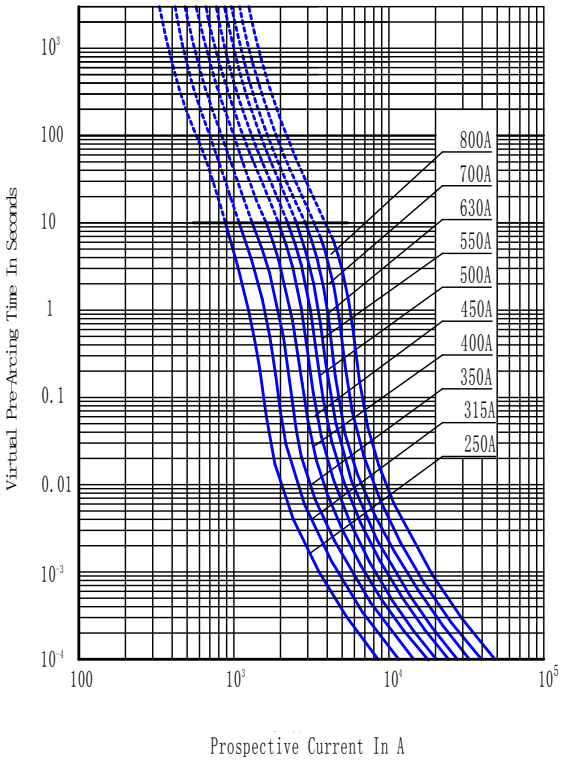
c）When the stud is screwed in, there is no recommended torque, and the depth of screw-in is 1mm below the thread

Nut	Nut tightening torque
M10	26± 1N.m

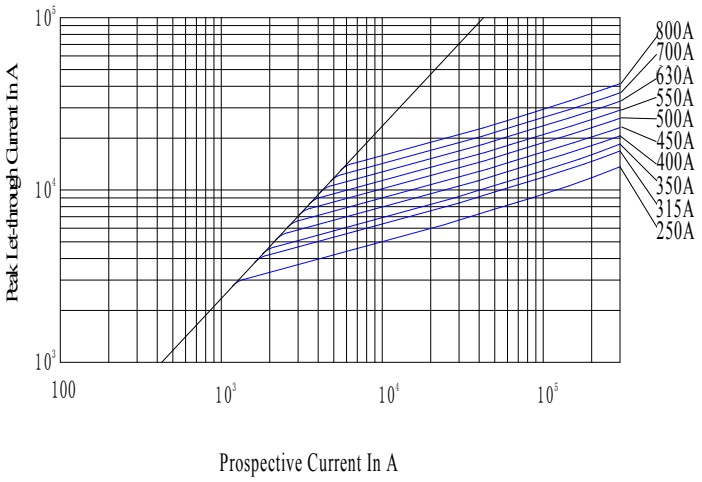
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Characteristic Curves:

Size2-Time Current Characteristic Curves:



Size2-Cut- off Current Characteristic Curves:



Note: 1. Curves: current tolerance ±10%;
2. Minimum breaking current at least $8I_n$.

Transport and Storage

Transport

Avoid rain/snow or mechanical damage during transportation.

Storage

Storage temp: -40°C ~ 120°C, Maximum 70% RH at 40°C;
Maximum 80% RH at 30°C; Maximum 90% RH at 20°C;
Package and Storage temp: -40°C～70°C, Maximum 90% RH ; no dewing.

Usage Conditions

Normal Condition and Corrections

Correction is not required under normal conditions.

For other conditions, if they are within tolerable range, certain correction measures may be required. If conditions are beyond tolerable range, please consult our team for evaluation and testing.

Long-term operation current is recommended to be <80% of rated current.

Ambient Temperature

Normal Condition
-5°C～40°C

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Tolerable Range

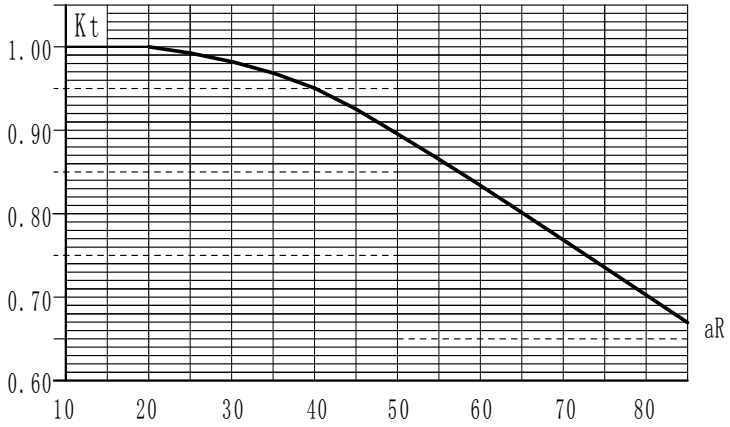
-40°C～85°C

Ambient temperature correction: operating below -5°C, resulting in longer pre-arc time under small overcurrent and slightly increased rated current.

If above 40°C, rated current is corrected as per factor -Kt.

Note 1: Kt value has considered safety margin of rated current during normal operation.

Note 2: ambient temperature should last 1-2 hrs. before it has a significant impact on fuse.



Altitude

Normal Condition

Below 2000m

Tolerable Condition

2000-4500m

Correction: higher altitude would affect insulation and dissipation, also changes air pressure.

a) For every 100m higher, fuse temperature rise increases by 0.1-0.5k.

b) For every 100m higher, ambient temperature drops by 0.5k approximately.

c) Normally for fuses in open environment, altitude condition is negligible.

d) For closed environment, if ambient temperature inside remains almost stable under different altitude.

If exceed 40°C, fuse should be degraded. For every 1000m, rated current should be degraded by 2%-5%.

Note: for same series, larger rated fuse should use higher degrade %, and lower degrade % for smaller one.

Air Insulation Strength (Breakdown)

a) Air insulation reduces with higher altitude. For 2000-4500m, insulation decreases by 12-15% for every 1000m as per GB/T16935.1. Thus adjust clearing space.

b) Space between fuse terminals is often much larger than specified value in standard.

c) User should consider altitude impact on spacing between fuse and other electric component, earthing etc.

Atmosphere

Normal Conditions

Clean atmosphere, maximum 50% RH at 40°C.

Higher RH is allowed when temperature is low, e.g. maximum 90% at 20 °C.

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Moderate dewing may occur under temperature changes.

Tolerable Conditions

If dewing is minor, RH could be up to 95%.

Vibration

It has great withstand to anti-vibration and mechanical shock, up to 10g acceleration.

Conform to rail transport vibration grade II.

The withstand to anti-vibration and mechanical shock conform to GB/T 28046.3 for elastic body of passenger car

For severe vibration application, please consult our team for evaluation and testing.

Pollution Class

Grade 3 pollution withstand

Mounting Condition

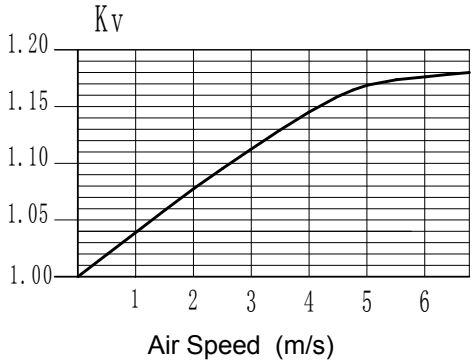
a) Installed in open air without any ventilation. No heat source within 1m except for conductors

b) Contact of fuses must be securely connected. Contact resistance should not affect operation.

c) Fuse can be mounted in any orientation. If spring compression is adopted, make sure it is properly mounted to avoid harmful effect due to gravity or vibration.

Forced Air and Liquid Cooling

Current carrying capacity of fuse can be improved by implementing forced air or liquid cooling.



Safety and Maintenance

a) Make sure sufficient clearance between installed fuses. Install insulation if necessary. This is to avoid possible inter-phase short circuit while replacing fuse.

b) Periodic maintenance per electric equipment. Remove oxidation, dusts on contacting part.

c) It is compulsory to replace all mechanically damaged fuses.

d) Unless permissive (e.g. fused load-switch), do not replace fuses while energized.

e) While servicing, fuse will not generate gas, dust, noise or others that may harm the environment

f) Metallic part of fuse can be recycled. Non-metal part can be crushed and treated as normal industry waste. It will not cause further pollution to the environment.